

**IDC vom Mainboard (Belegung Spec 2.4)**

The schematic shows the following components and their connections:

- MCP23017 (0x20):** Taster 1–10 gegen GND (interne Pull-ups), Encoder, DISP\_RST (10k Pull-up, geht ueber IDC 29 auch zu den Augen), Backlight-Enable; INTA direkt an IO\_INT.
- Bauch-Display ILI9341 2.8in SPI (MSP2807, 14-Pin, Header links):** Backlight 150 mA; High-Side BC327 direkt vom MCP GPB6 (low = an); 10k B-E haelt aus, bis der MCP konfiguriert ist.
- Fader X32 (panel-mount, Draehete an Stifteleisten):** Motor auf IDC, Schleifer ueber RC 1k/100n, Touch ueber J18 zum MPR121 (Modul per Kabel, kein Sockel; Clone-Raster unbelagt, Falle 15).
- Mechanik:** 4 Ecken + 4 Display-Abstandshalter (M3 x 11 mm, Raster 76,08 x 44).
- PWR\_FLAGS:** Three 3V3 voltage regulators connected to GND.

**Main Board Connections (IDC 2x15):**

- SW1, SW2, SW3, SW4, SW5, SW6, SW7, SW8, SW9, SW10
- GND, 3V3, MOT1A, MOT1B, MOT2A, MOT2B, MOT3A, MOT3B, MOT4A, MOT4B
- FADER1\_WIPER, FADER2\_WIPER, FADER3\_WIPER, FADER4\_WIPER
- I2C\_SCL, I2C\_SDA, IO\_INT, CS\_BELLY, SPL\_MOSI, SPL\_SCK, DISP\_RST

**Other Components:**

- ENC\_A, ENC\_B, ENC\_SW
- BC327 (High-Side Switch)
- J14, J15, J16, J17 (Motor headers)
- J18 (Touch header)
- MPR121 (Module)